Project Output

Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

= RESTART: E:\Real Lif Project In Data Analyst\Job-Oreiented Data Science Project Scenario\employee_data.py

==========

employee_data.csv Created Successfully!

Total Records: 1000

==========

==========Practical Task List==========

	Employee ID	Employee Name	Department	...	Training Hours	Joining Date	Attrition
0	C001	Amit	Finance	...	32	24-08-2026	No
1	C002	Swati	Sales	...	46	15-05-2026	No
2	C003	Swati	IT	...	25	15-12-2026	No
3	C004	Akash	Finance	...	48	07-11-2026	No
4	C005	Kiran	HR	...	15	04-01-2026	Yes
..	...	...	...	...	...	...	...
994	C995	Priya	HR	...	19	10-12-2026	No
995	C996	Swati	Sales	...	26	17-12-2026	No
996	C997	Deepak	Finance	...	38	06-02-2026	Yes
997	C998	Anajali	Sales	...	38	22-12-2026	No
998	C999	Deepak	Finance	...	49	17-12-2026	No

[999 rows x 13 columns]

======First 10 and Last 5 Records=======

First 10: Employee ID Employee Name Department ... Training Hours Joining Date Attrition

0 C001 Amit Finance ... 32 24-08-2026 No

1 C002 Swati Sales ... 46 15-05-2026 No

2 C003 Swati IT ... 25 15-12-2026 No

3 C004 Akash Finance ... 48 07-11-2026 No

4 C005 Kiran HR ... 15 04-01-2026 Yes

5 C006 Priya Sales ... 41 04-01-2026 Yes

6 C007 Anajali Marketing ... 37 14-07-2026 Yes

7 C008 Rahul Sales ... 28 31-12-2026 Yes

8 C009 Neha Finance ... 29 23-11-2026 No

9 C010 Vikas Marketing ... 23 22-02-2026 No

[10 rows x 13 columns]

Last 5: Employee ID Employee Name Department ... Training Hours Joining Date Attrition

994 C995 Priya HR ... 19 10-12-2026 No

995 C996 Swati Sales ... 26 17-12-2026 No

996 C997 Deepak Finance ... 38 06-02-2026 Yes

997 C998 Anajali Sales ... 38 22-12-2026 No

998 C999 Deepak Finance ... 49 17-12-2026 No

[5 rows x 13 columns]

======Find The Number Of Rows & Columns======

Number Of Rows & Columns: (999, 13)

======Column Names And Data Types======

Columns Names: Index(['Employee ID', 'Employee Name', 'Department', 'Job Role', 'Age',

'Gender', 'Experience Years', 'Monthly Salary', 'Performance Score',

'Project Completed', 'Training Hours', 'Joining Date', 'Attrition'],

dtype='str')

Data Types: Employee ID str

Employee Name str

Department str

Job Role str

Age int64

Gender str

Experience Years int64

Monthly Salary int64

Performance Score int64

Project Completed int64

Training Hours int64

Joining Date str

Attrition str

dtype: object

======Statistical Summary Describe======

Describe:	Age	Experience Years	...	Project Completed	Training Hours
count	999.000000	999.000000	...	999.000000	999.000000
mean	42.772773	6.462462	...	8.194194	30.185185
std	13.261642	3.480023	...	4.254471	12.079759
min	21.000000	1.000000	...	1.000000	10.000000
25%	31.000000	3.000000	...	5.000000	19.500000
50%	43.000000	7.000000	...	8.000000	31.000000
75%	54.000000	10.000000	...	12.000000	40.500000
max	65.000000	12.000000	...	15.000000	50.000000

[8 rows x 6 columns]

==========Data Cleaning==========

======Remove Duplicate Records======

Remove Duplicate: 0

======Missing Values In Each Column======

Missing Values: Employee ID 0

Employee Name 0

Department 0

Job Role 0

Age 0

Gender 0

Experience Years 0

Monthly Salary 0

Performance Score 0

Project Completed 0

Training Hours 0

Joining Date 0

Attrition 0

dtype: int64

======Missing Salary Values With The Departments Median Salary======

Missing Salary Values: 0 32000

1 45000

2 22000

3 44000

4 32000

...

994 32000

995   49000

996   35000

997   25000

998   32000

Name: Monthly Salary, Length: 999, dtype: int64

======Inconsistent Department Names And Extra Space======

<StringArray>

['Finance', 'Sales', 'IT', 'HR', 'Marketing']

Length: 5, dtype: str

======Joining Date Into Datetime Formate======

0   2026-08-24

1   2026-05-15

2   2026-12-15

3   2026-11-07

4   2026-01-04

       ...

994   2026-12-10

995   2026-12-17

996   2026-02-06

997   2026-12-22

998   2026-12-17

Name: Joining Date, Length: 999, dtype: datetime64[us]

======Negative Salaries And Invalid Performance Scores======

Monthly Salary   0

Performance Score   0

dtype: int64

Empty DataFrame

Columns: [Employee ID, Employee Name, Department, Job Role, Age, Gender, Experience Years, Monthly Salary, Performance Score, Project Completed, Training Hours, Joining Date, Attrition]

Index: []

==========Data Transformation==========

======Create Annual Salary======

	Employee Name	Monthly Salary	Annual Salary
0	Amit	32000	384000
1	Swati	45000	540000
2	Swati	22000	264000
3	Akash	44000	528000
4	Kiran	32000	384000
..	...	...	...
994	Priya	32000	384000
995	Swati	49000	588000
996	Deepak	35000	420000
997	Anajali	25000	300000
998	Deepak	32000	384000

[999 rows x 3 columns]

======Experience Level======

	Employee ID	Employee Name	...	Annual Salary	Experience Level
0	C001	Amit	...	384000	Fresher
1	C002	Swati	...	540000	Fresher
2	C003	Swati	...	264000	Mid-Level
3	C004	Akash	...	528000	Senior
4	C005	Kiran	...	384000	Mid-Level
..	...	...	...	...	...

994 C995 Priya ... 384000 Mid-Level

995 C996 Swati ... 588000 Mid-Level

996 C997 Deepak ... 420000 Senior

997 C998 Anajali ... 300000 Senior

998 C999 Deepak ... 384000 Senior

[999 rows x 15 columns]

======Performance Category======

Employee ID Employee Name ... Experience Level Performance Category

0 C001 Amit ... Fresher High

1 C002 Swati ... Fresher Average

2 C003 Swati ... Mid-Level Low

3 C004 Akash ... Senior High

4 C005 Kiran ... Mid-Level High

.. ..

994 C995 Priya ... Mid-Level Average

995 C996 Swati ... Mid-Level High

996 C997 Deepak ... Senior High

997 C998 Anajali ... Senior Average

998 C999 Deepak ... Senior Average

[999 rows x 16 columns]

======Salary Band=======

Employee ID Employee Name ... Performance Category Salary Band

0 C001 Amit ... High High

1 C002 Swati ... Average High

2 C003 Swati ... Low Medium

3 C004 Akash ... High High

4 C005 Kiran ... High High

.. ..

994 C995 Priya ... Average High

995 C996 Swati ... High High

996 C997 Deepak ... High High

997 C998 Anajali ... Average High

998 C999 Deepak ... Average High

[999 rows x 17 columns]

======Years In Company From The Joining Date=======

year Joining Date

0 2026 2026-08-24

1 2026 2026-05-15

2 2026 2026-12-15

3 2026 2026-11-07

4 2026 2026-01-04

.. ..

994 2026 2026-12-10

995 2026 2026-12-17

996 2026 2026-02-06

997 2026 2026-12-22

998 2026 2026-12-17

[999 rows x 2 columns]

==========Business Analysis==========

=======Average Salary By Department=======

Department

Sales 37593.750000

Finance 37118.226601

Marketing 37038.834951

IT 36575.268817

HR 36122.641509

Name: Monthly Salary, dtype: float64

=======Highest Paid Employee=======

Employee Name

Deepak 2284000

Neha 2123000

Akash 2073000

Riya 2012000

Amit 2010000

Name: Monthly Salary, dtype: int64

=======Top 10 Employees By Performance Score=======

Employee Name

Akash 336

Amit 321

Neha 317

Sachin 306

Rahul 302

Nitin 297

Deepak 295

Riya 284

Vikas 283

Priya 273

Name: Performance Score, dtype: int64

=======Average Salary By Job Role=======

Job Role

Data Cleaning 37776.315789

Operator 37496.894410

HR Executive 37197.674419

Developer 36843.930636

Data Science 36512.195122

Data Analyst 35621.468927

Name: Monthly Salary, dtype: float64

=======Average Experience Across Department=======

Department

Finance 6.733990

Marketing 6.660194

HR 6.575472

Sales 6.265625

IT 6.021505

Name: Experience Years, dtype: float64

=======Employee Attrition Percentage=======

Total Employee: 999

Employee Left: 479

Attrition Percentage:47.95%

=======Employee With High Salaries But Low Performance=======

	Employee ID	Employee Name	Department	...	Performance Category	Salary Band	year
0	C001	Amit	Finance	...	High	High	2026
1	C002	Swati	Sales	...	Average	High	2026
3	C004	Akash	Finance	...	High	High	2026
4	C005	Kiran	HR	...	High	High	2026
5	C006	Priya	Sales	...	High	High	2026

.. ..

994 C995 Priya HR ... Average High 2026

995 C996 Swati Sales ... High High 2026

996 C997 Deepak Finance ... High High 2026

997 C998 Anajali Sales ... Average High 2026

998 C999 Deepak Finance ... Average High 2026

[863 rows x 18 columns] Employee ID Employee Name Department ... Performance Category Salary Band year

2 C003 Swati IT ... Low Medium 2026

7 C008 Rahul Sales ... Low Medium 2026

16 C017 Rohit Finance ... Low Medium 2026

17 C018 Pooja HR ... Low High 2026

32 C033 Deepak Finance ... Low High 2026

.. ..

957 C958 Deepak Finance ... Low High 2026

979 C980 Rakesh HR ... Low High 2026

980 C981 Riya IT ... Low High 2026

981 C982 Vikas Marketing ... Low High 2026

989 C990 Rohit IT ... Low Medium 2026

[106 rows x 18 columns]

==========Advanced Pandas==========

=======Summarize Department Performance=======

count mean min max

Department

Finance 203 5.177340 1 10

HR 212 5.745283 1 10

IT 186 5.607527 1 10

Marketing 206 5.368932 1 10

Sales 192 5.442708 1 10

=======Multiple Statistics=======

count mean min max sum median std

Department

Finance 203 5.177340 1 10 1051 5.0 2.854227

HR 212 5.745283 1 10 1218 6.0 2.961239

IT 186 5.607527 1 10 1043 6.0 2.913482

Marketing 206 5.368932 1 10 1106 5.0 2.876279

Sales 192 5.442708 1 10 1045 5.0 2.866460

=======Sort Values To Rank Employees=======

Employee ID Employee Name Department ... Salary Band year Rank

498 C499 Komal Marketing ... Medium 2026 1

48 C049 Anajali IT ... High 2026 2

954 C955 Nitin Sales ... Medium 2026 3

481 C482 Pooja Sales ... High 2026 4

479 C480 Rahul Sales ... High 2026 5

.. ..

7 C008 Rahul Sales ... Medium 2026 995

980 C981 Riya IT ... High 2026 996

979 C980 Rakesh HR ... High 2026 997

2 C003 Swati IT ... Medium 2026 998

17 C018 Pooja HR ... High 2026 999

[999 rows x 19 columns]

=======Loc And I Loc to Filter Records=======

Empty DataFrame

Columns: [Employee ID, Employee Name, Department, Job Role, Age, Gender, Experience Years, Monthly Salary, Performance Score, Project Completed, Training Hours, Joining Date, Attrition, Annual Salary, Experience Level, Performance Category, Salary Band, year, Rank]

Index: []

	Employee ID	Employee Name	Department	...	Salary Band	year	Rank
656	C657	Kiran	Finance	...	High	2026	7
61	C062	Komal	Finance	...	High	2026	11
509	C510	Riya	Finance	...	High	2026	13
677	C678	Rahul	Finance	...	High	2026	17
717	C718	Anajali	Finance	...	High	2026	44
..	...	...	...	...	...	...	...
687	C688	Riya	Finance	...	High	2026	988
32	C033	Deepak	Finance	...	High	2026	990
577	C578	Riya	Finance	...	High	2026	991
16	C017	Rohit	Finance	...	Medium	2026	992
566	C567	Sagar	Finance	...	High	2026	994

[203 rows x 19 columns]

	Employee ID	Employee Name	Department	...	Salary Band	year	Rank
954	C955	Nitin	Sales	...	Medium	2026	3
567	C568	Sagar	Marketing	...	High	2026	6
61	C062	Komal	Finance	...	High	2026	11
94	C095	Priya	IT	...	High	2026	15
677	C678	Rahul	Finance	...	High	2026	17
..	...	...	...	...	...	...	...
372	C373	Swati	HR	...	High	2026	976
719	C720	Anajali	Marketing	...	High	2026	977
84	C085	Sachin	IT	...	Medium	2026	985

407 C408 Meena Marketing ... Medium 2026 986

434 C435 Anajali Marketing ... High 2026 987

[177 rows x 19 columns]

Employee ID C482

Employee Name Pooja

Department Sales

Job Role Data Cleaning

Age 65

Gender Male

Experience Years 12

Monthly Salary 43000

Performance Score 10

Project Completed 12

Training Hours 48

Joining Date 2026-01-19 00:00:00

Attrition Yes

Annual Salary 516000

Experience Level Senior

Performance Category High

Salary Band High

year 2026

Rank 4

Name: 481, dtype: object

Employee ID C004

Employee Name Akash

Department Finance

Job Role Data Cleaning

Age 59

Gender Female

Experience Years 11

Monthly Salary 44000

Performance Score 7

Project Completed 1

Training Hours 48

Joining Date 2026-11-07 00:00:00

Attrition No

Annual Salary 528000

Experience Level Senior

Performance Category High

Salary Band High

year 2026

Rank 410

Name: 3, dtype: object

Empty DataFrame

Columns: []

Index: [498, 48, 954]

=======Pivot Table Compare Salary And Performance=======

Monthly Salary

Performance Score

1 3617000

2 3675000

3 3638000

4 4260000

5 3338000

6 3265000

7 4045000

8 3977000

9 3227000

10 3802000

=======Merge To Combine Employee And Department Budget Datasets=======

Department Department_Budget

0 IT 25000

1 HR 23000

2 Sales 45000

3 Finance 26000

4 Marketing 55000

<bound method DataFrame.merge of Employee ID Employee Name Department ... year Rank Department_Budget

0 C499 Komal Marketing ... 2026 1 55000

1 C049 Anajali IT ... 2026 2 25000

2 C955 Nitin Sales ... 2026 3 45000

3 C482 Pooja Sales ... 2026 4 45000

4 C480 Rahul Sales ... 2026 5 45000

.. ..

994 C008 Rahul Sales ... 2026 995 45000

995 C981 Riya IT ... 2026 996 25000

996 C980 Rakesh HR ... 2026 997 23000

997 C003 Swati IT ... 2026 998 25000

998 C018 Pooja HR ... 2026 999 23000

[999 rows x 20 columns]>

=======Apply Vectorized Operations To Classify Employees=======

Employee ID Employee Name Department ... Salary Band year Rank

498 Excellent Komal Marketing ... Medium 2026 1

48 Excellent Anajali IT ... High 2026 2

954 Excellent Nitin Sales ... Medium 2026 3

481 Excellent Pooja Sales ... High 2026 4

479 Excellent Rahul Sales ... High 2026 5

.. ..

7 Needs Improvement Rahul Sales ... Medium 2026 995

980 Needs Improvement Riya IT ... High 2026 996

979 Needs Improvement Rakesh HR ... High 2026 997

2 Needs Improvement Swati IT ... Medium 2026 998

17 Needs Improvement Pooja HR ... High 2026 999

[999 rows x 19 columns]

==========Business Deliverables==========

=======Cleaned Dataset To Cleaned Employee Data.xlsx=======

=======Department Summaries To A Separated Excel Sheet=======

Department

HR 212

Marketing 206

Finance 203

Sales 192

IT 186

Name: Monthly Salary, dtype: int64 Department

Sales 37593.750000

Finance 37118.226601

Marketing 37038.834951

IT 36575.268817

HR 36122.641509

Name: Monthly Salary, dtype: float64 Department

HR 7658000

Marketing 7630000

Finance 7535000

Sales 7218000

IT 6803000

Name: Monthly Salary, dtype: int64 Department

HR 212

Marketing 206

Finance 203

Sales 192

IT 186

Name: Performance Score, dtype: int64 Department

HR 212

Marketing 206

Finance 203

Sales 192

IT 186

Name: Gender, dtype: int64

==========All Department Summaries Export Successfully==========

=======Top 10 Performance In A Separate Sheet=======

	Employee ID	Employee Name	Department	...	Salary Band	year	Rank
498	Excellent	Komal	Marketing	...	Medium	2026	1
48	Excellent	Anajali	IT	...	High	2026	2
954	Excellent	Nitin	Sales	...	Medium	2026	3
481	Excellent	Pooja	Sales	...	High	2026	4

479 Excellent Rahul Sales ... High 2026 5

567 Excellent Sagar Marketing ... High 2026 6

656 Excellent Kiran Finance ... High 2026 7

950 Excellent Amit IT ... High 2026 8

457 Excellent Sagar IT ... High 2026 9

658 Excellent Meena HR ... High 2026 10

[10 rows x 19 columns]

==========Top 10 Performance Export Successfully==========

======Fianl Report Containing At Least 10 Business insights=======

======Business Insights Report=======

Total Department: 999

Employee Per Department: Department

HR 212

Marketing 206

Finance 203

Sales 192

IT 186

Name: Employee ID, dtype: int64

Average Monthly Salary: 36880.88088088088

Highest Paid Employee: Employee Name

Deepak 2284000

Neha 2123000

Akash 2073000

Riya 2012000

Amit 2010000

Nitin 1959000

Priya 1940000

Vikas 1927000

Sachin 1923000

Rahul 1909000

Name: Monthly Salary, dtype: int64

Lowest Paid Employee: Employee Name

Rohit 1419000

Swati 1499000

Komal 1507000

Rakesh 1628000

Kiran 1672000

Meena 1687000

Sneha 1704000

Pooja 1791000

Anajali 1872000

Sagar 1905000

Name: Monthly Salary, dtype: int64

Average Performance Employee: Employee ID

Excellent 8.976667

Average 6.080986

Needs Improvement 2.513253

Name: Performance Score, dtype: float64

Top Performing Employees: Employee ID

Excellent 2693

Average 1727

Needs Improvement 1043

Name: Performance Score, dtype: int64

Department Average Performance: Department

HR 5.745283

IT 5.607527

Sales 5.442708

Marketing 5.368932

Finance 5.177340

Name: Performance Score, dtype: float64

Salary Expense: Department

HR 7658000

Marketing 7630000

Finance 7535000

Sales 7218000

IT 6803000

Name: Monthly Salary, dtype: int64

Department Budget: Department Department_Budget

0 IT 25000

1 HR 23000

2 Sales 45000

3 Finance 26000

4 Marketing 55000

Department Wise Gender: Department

HR 212

Marketing 206

Finance 203

Sales 192

IT 186

Name: Gender, dtype: int64

Job Role Monthly Total Salary: Job Role

HR Executive 6398000

Developer 6374000

Data Analyst 6305000

Operator 6037000

Data Science 5988000

Data Cleaning 5742000

Name: Monthly Salary, dtype: int64

==========Business Questions==========

=======Department Has The Highest Average Salary=======

Department

Sales 37593.750000

Finance 37118.226601

Marketing 37038.834951

IT 36575.268817

HR 36122.641509

Name: Monthly Salary, dtype: float64

=======Job Role Has The Highest Salary=======

Job Role

HR Executive 6398000

Developer 6374000

Data Analyst 6305000

Operator 6037000

Data Science 5988000

Data Cleaning 5742000

Name: Monthly Salary, dtype: int64

=======Employee With More Experience Have Higher Salaries=======

Experience Years

2 38943.661972

9 38487.179487

11 37825.581395

12 37600.000000

3 37022.222222

1 36918.367347

4 36730.769231

8 36511.627907

7 36279.069767

6 35848.484848

5 35677.777778

10 35284.210526

Name: Monthly Salary, dtype: float64

=======Which Department Has The Highest Attrition Rate=======

Department

HR 212

Marketing 206

Finance 203

Sales 192

IT 186

Name: Attrition, dtype: int64

=======Employees Have a Performance Score Above 8 & Experience Below 3 Years=======

	Employee ID	Employee Name	Department	...	Salary Band	year	Rank
48	Excellent	Anajali	IT	...	High	2026	2
515	Excellent	Deepak	Sales	...	High	2026	12
689	Excellent	Komal	HR	...	High	2026	23
939	Excellent	Neha	Marketing	...	High	2026	26
892	Excellent	Meena	IT	...	High	2026	34
344	Excellent	Rohit	IT	...	Medium	2026	36

163 Excellent Deepak IT ... High 2026 40

717 Excellent Anajali Finance ... High 2026 44

381 Excellent Amit Marketing ... High 2026 53

995 Excellent Swati Sales ... High 2026 64

804 Excellent Anajali IT ... High 2026 71

595 Excellent Deepak IT ... High 2026 74

194 Excellent Rohit Sales ... High 2026 78

269 Excellent Rahul Marketing ... High 2026 82

874 Excellent Riya IT ... High 2026 89

181 Excellent Sagar HR ... High 2026 92

182 Excellent Amit HR ... High 2026 93

992 Excellent Amit Finance ... High 2026 95

811 Excellent Nitin Sales ... High 2026 99

766 Excellent Rakesh IT ... High 2026 102

477 Excellent Neha Finance ... High 2026 113

651 Excellent Rahul Finance ... High 2026 117

473 Excellent Sneha HR ... High 2026 118

55 Excellent Sneha Marketing ... Medium 2026 125

91 Excellent Nitin HR ... High 2026 130

312 Excellent Rakesh Sales ... High 2026 133

171 Excellent Swati HR ... High 2026 134

353 Excellent Neha HR ... High 2026 140

738 Excellent Sneha HR ... Medium 2026 144

699 Excellent Swati Finance ... High 2026 150

117 Excellent Nitin Finance ... High 2026 157

159 Excellent Pooja HR ... High 2026 162

560 Excellent Pooja Sales ... Medium 2026 167

871 Excellent Meena Marketing ... High 2026 169

782 Excellent Pooja IT ... High 2026 174

843 Excellent Amit Sales ... High 2026 183

839 Excellent Rohit HR ... High 2026 187

978 Excellent Pooja HR ... Medium 2026 191

[38 rows x 19 columns]

=======Percentage of Employees Earn More Than 50000 Per Month======

Employee Earn 50000: 112

Total Employee: 999

Percentage Employee: 11.21 %

=======Department has The Highest Average Performance Score=======

Department

HR 5.745283

IT 5.607527

Sales 5.442708

Marketing 5.368932

Finance 5.177340

Name: Performance Score, dtype: float64

=======Employee Completing More Projects Associated With higher Performance Score=======

Project Completed

8 79

14 73

10 72

11 70

4 70

5 70

12 70

2 68

9 68

15 67

6 65

13 63

7 57

1 55

3 52

Name: Performance Score, dtype: int64

=======Employees May Require Performance Improvement Plans=======

Employee ID Employee Name Department ... Salary Band year Rank

971 Needs Improvement Deepak HR ... High 2026 585

969 Needs Improvement Kiran Finance ... Medium 2026 586

9 Needs Improvement Vikas Marketing ... High 2026 587

11 Needs Improvement Kiran HR ... High 2026 588

966 Needs Improvement Anajali Sales ... High 2026 589

.. ..

7 Needs Improvement Rahul Sales ... Medium 2026 995

980 Needs Improvement Riya IT ... High 2026 996

979 Needs Improvement Rakesh HR ... High 2026 997

2 Needs Improvement Swati IT ... Medium 2026 998

17 Needs Improvement Pooja HR ... High 2026 999

[415 rows x 19 columns]

=======HR Manager Based On The Findings=======

Identify employee with low performance scores

Recognize and reward high-performing employees

Analyze the relationship between experience

Review employees earning above 50000 per month

==========End Of Recomemendations==========